

LifeLink

Blood Donation and Emergency Blood Matching System

Project Report

1. Abstract

LifeLink is a database-driven emergency blood matching platform designed to help hospitals quickly find compatible and available blood resources. By considering blood group compatibility, donor location, availability, and recent donation history, the system eliminates delays in critical situations.

Key Focus

The platform guarantees data integrity and reliable patient care by maintaining a highly structured and optimized matching process, streamlining the connection between donors and critical emergency needs.

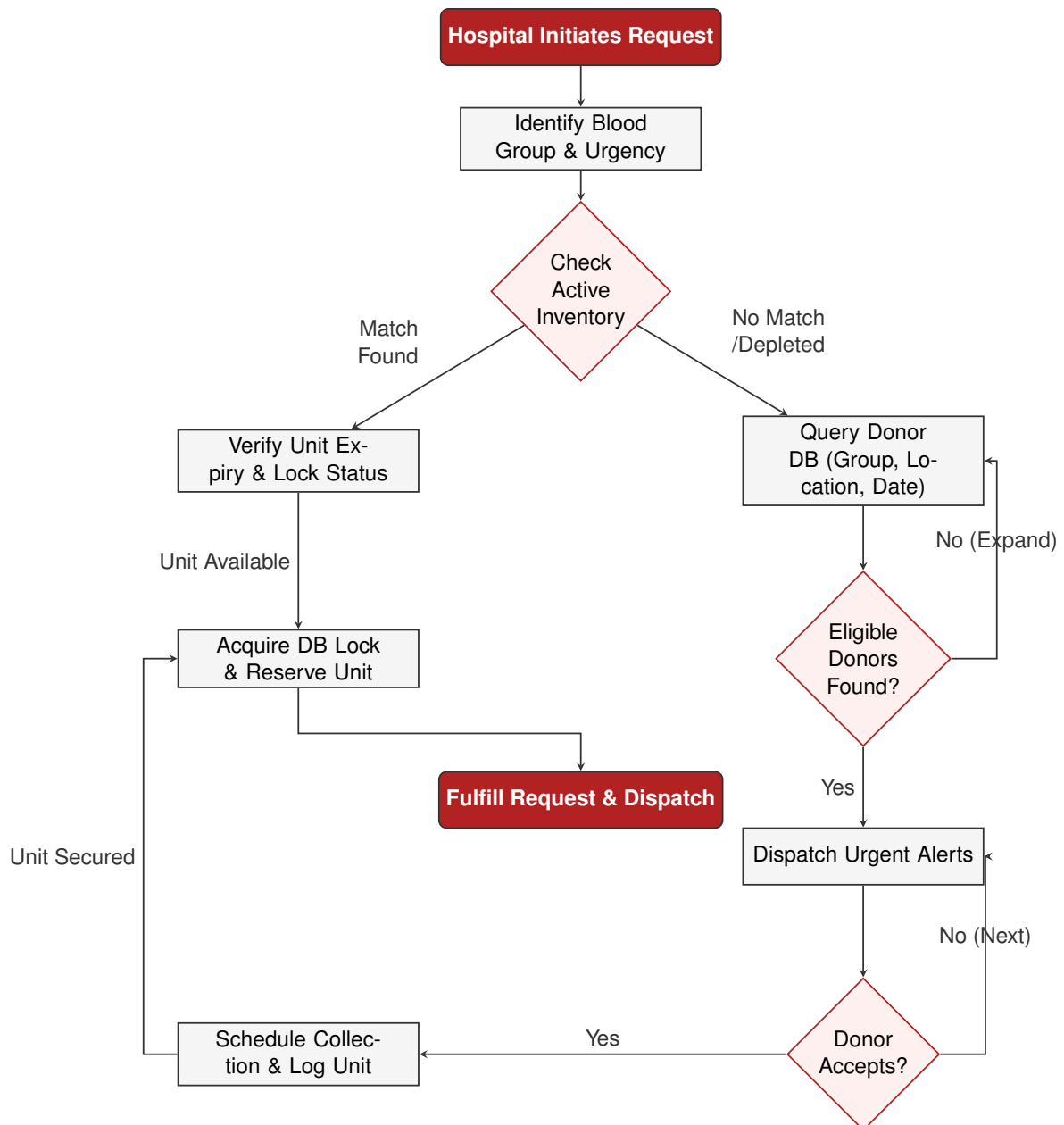
2. Problem Statement

During medical emergencies, hospitals and patients often face life-threatening delays in securing specific blood types. The challenge is not merely storing a list of donors, but rapidly filtering that list to find donors or existing blood units that meet a strict set of real-time criteria:

- **Medical Compatibility:** Matching exact or compatible complex blood groups (e.g., O-universally, AB+ receiving).
- **Geographic Proximity:** Ensuring the donor or blood unit is near the emergency location to minimize transit time.
- **Availability & Eligibility:** Verifying the donor has not donated too recently (maintaining the mandated cool-down period) and is actively available.

3. Detailed System Workflow

The system relies on a multi-tiered filtering mechanism. When a request is triggered, it first queries the existing on-hand inventory. If the inventory is insufficient, the algorithm seamlessly transitions to querying the live donor database, filtering by medical eligibility and geographic distance.



Workflow Breakdown:

- 1. Request Initiation:** Hospital submits a critical request detailing required blood type and urgency.
- 2. Inventory Query (Fast Path):** The system scans BLOOD INVENTORY. If a match is found, it is verified and locked to prevent concurrent reservation conflicts.
- 3. Donor Search (Fallback Path):** If inventory is depleted, the system queries the DONOR entity using strict filters (compatibility, distance, cooldown period).
- 4. Notification & Response:** Eligible donors receive immediate alerts. The first acceptance reserves the slot.
- 5. Fulfillment & Logging:** Blood is collected/retrieved, transaction commits, inventory updates, and hospital is notified.

4. Database Design & Entities

The core of LifeLink relies on a highly normalized relational database to maintain data integrity and support the complex filtering logic outlined in the workflow.

Core Entities:

- **DONOR:** Donor_ID (PK), Name, Blood_Group, Last_Donation_Date, Location_Coordinates, Contact, Is_Available.
- **HOSPITAL / PATIENT:** Hospital_ID (PK), Name, Location_Coordinates, Emergency_Contact.
- **BLOOD INVENTORY:** Unit_ID (PK), Blood_Group, Collection_Date, Expiry_Date, Status (Available / Locked / Used).
- **BLOOD REQUEST:** Request_ID (PK), Hospital_ID (FK), Required_Blood_Group, Urgency_Level, Status.
- **DONATION HISTORY:** Donation_ID (PK), Donor_ID (FK), Date, Unit_ID (FK).

5. Conclusion

The **LifeLink** project moves beyond basic data storage by solving a time-sensitive, real-world logistical problem. It successfully balances simplicity in its workflow with advanced structural design. The result is a robust, socially impactful database application that addresses critical healthcare resource management efficiently while demonstrating strong relational database management principles.